

1THE PROBLEM: SAFE SG → GFL REPLACEMENT

Replacing a synchronous generator (SG) by a grid-following inverter (GFL) is a **local** hardware action, but its dynamical consequence is **collective**. The admissibility of a replacement belongs to the closed grid, not to the device in isolation.

$$\delta_i = \begin{cases} 0, & \text{retain SG}_i, \\ 1, & \text{SG}_i \rightarrow \text{GFL}_i \end{cases}$$

$$S(\delta) = \{ i : \delta_i = 1 \}$$

P_i^0 rated/scheduled active generation
 K GFL controller architecture and gains
 $\alpha_{\perp}$ transverse spectral abscissa

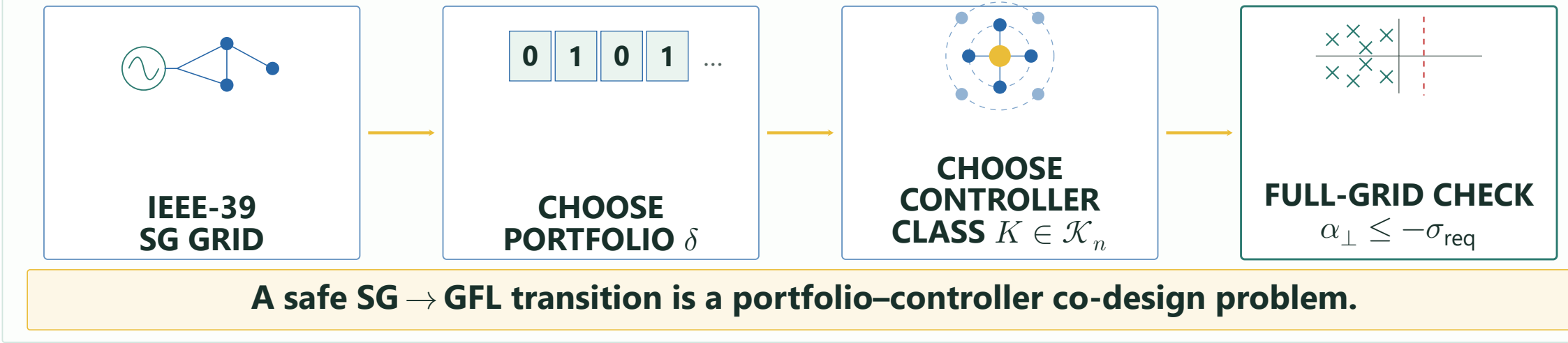
DESIGN OBJECTIVE

$$\max_{S, K \in \mathcal{K}_n} \sum_{i \in S} P_i^0$$

subject to $\alpha_{\perp}(S, K) \leq -\sigma_{\text{req}}$

$K \in \mathcal{K}_n$, $K^- \leq K \leq K^+$, physical/event guards satisfied.

The paper reports the largest verified design found; global optimality is not claimed.



4EXACT NETWORK CLOSURE: LOCAL × COLLECTIVE

2MODEL: DAE → TRANSVERSE STABILITY

3IEEE-39: MINIMAL COLLECTIVE BLOCKER

7INFORMATION RADIUS AND N-HOP CONTROL

5BEYOND NODAL DAMPING: EFFECTIVE OPERATORS

6GRID-EMBEDDED PLL GAIN SYNTHESIS

MAIN CONTRIBUTIONS

SCOPE OF THE RESULTS

REFERENCES / CONTACT

PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

More information gives more control authority — only the full grid decides whether a larger SG → GFL Transition is safe.